

1. Raising the search depth of alpha-beta enhances move quality, as it allows the agent to anticipate further moves and prevent basic tactical errors. Shallow depths frequently miss multi-jumps or promotions, whereas medium depths offer significantly more stable play. Improving the evaluation function by assigning greater value to kings, mobility, and board position also boosts the performance of an agent. This is particularly true in late-game scenarios, where simply counting pieces is insufficient.
2. Using the suggested constant $C = \sqrt{2}$ offers the optimal equilibrium between exploration and exploitation in MCTS. If C is minor, it makes the agent too greedy and less capable of finding better moves. On the other hand, if C is larger, it leads to excessive random exploration and weaker choices. In general terms, $C = \sqrt{2}$ resulted in the most stable outcomes.
3. The alpha-beta tool demonstrated the best overall performance due to its dependable and specific search capabilities. MCTS performed exceptionally well in complicated situations, but not always consistently, as it relies on random simulations. The hybrid tool yielded variable results, as it randomly alternates between these two strategies. Performance ranking in general: Alpha-Beta is greater than MCTS, which is greater than Hybrid.